

Definition

Let X be a random variable.

If X is a DRV with pmf $p(x)$ and $\sum_x |x|p(x) < \infty$, then the expected value of X is

$$E(X) = \sum_x xp(x)$$

If X is a CRV with pdf $f(x)$ and $\int_{-\infty}^{\infty} |x|f(x)dx < \infty$, then the expected value of X is

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

- We use $|x|$ to ensure absolute convergence
- We often use μ to denote $E(X)$
- Sometimes $E(X)$ is called the expected value of X , or the weighted mean of X

Theorem

Let X be a random variable, and let $Y = g(X)$ for some function g .

Suppose X is discrete with pmf $p_x(x)$ and support S_x . If $\sum_{x \in S_x} |g(x)|p_x(x) < \infty$, then the expectation of Y exists and is given by

$$E(Y) = \sum_{x \in S_x} g(x)p_x(x)$$

Suppose X is continuous with pdf $f_x(x)$. If $\int_{-\infty}^{\infty} |g(x)|f_x(x)dx < \infty$, then the expectation of Y exists and is given by

$$E(Y) = \int_{-\infty}^{\infty} g(x)f_x(x)dx$$

Example

Let X be a DRV with pmf $p(x) = \left(\frac{1}{2}\right)^{x+1}$ for $x = 0, 1, 2, \dots$ and $Y = e^{-X}$. Find $E(Y)$

$$\Rightarrow E(Y) = E(e^{-X}) = \sum_{x=0}^{\infty} e^{-x} \left(\frac{1}{2}\right)^{x+1} = \frac{1}{2} \sum_{x=0}^{\infty} \left(\frac{1}{2}\right)^x = \frac{1}{2} \sum_{x=0}^{\infty} \left(\frac{1}{2}\right)^x = \frac{1}{2} \left(\frac{1}{1-\frac{1}{2}}\right) = \frac{1}{2} \left(\frac{2}{2-1}\right) = \frac{1}{2} \left(\frac{2}{1}\right) = \frac{1}{1} = 1$$

Example

Let X be a CRV with pdf $f(x) = 2x$ for $0 < x < 1$ and $Y = \frac{1}{1+x}$. Find $E(Y)$

$$\begin{aligned} \Rightarrow E(Y) &= E\left(\frac{1}{1+x}\right) = \int_0^1 \left(\frac{1}{1+x}\right)(2x)dx = 2 \int_0^1 \frac{x}{1+x} dx = 2 \int_0^1 \frac{x+1-1}{1+x} dx \quad (\text{where } u = 1+x) \\ &= 2 \int_0^1 \left(1 - \frac{1}{1+x}\right) dx = 2(1) - 2 \left[\ln(1+x)\right]_0^1 \\ &= 2(1 - (\ln 2 - \ln 1)) \\ &= 2(1 - \ln 2) \end{aligned}$$

Linearity

Let $g_1(x)$ and $g_2(x)$ be functions of a random variable X . Suppose the expectations of $g_1(x)$ and $g_2(x)$ exist.

Then for any constants k_1 and k_2 , the expectation of $k_1g_1(x) + k_2g_2(x)$ exists and is given by

$$E(k_1g_1(x) + k_2g_2(x)) = k_1E(g_1(x)) + k_2E(g_2(x))$$

Example

Let X have the pmf $p(x) = \frac{x}{6}$ for $x = 1, 2, 3$, 0 elsewhere. Find $E(6X^2 + X)$

$$\Rightarrow E(6X^2 + X) = 6E(X^2) + E(X) = \sum_{x=1}^3 x^2 p(x) + \sum_{x=1}^3 x p(x) = \sum_{x=1}^3 x^2 \left(\frac{x}{6}\right) + \sum_{x=1}^3 x \left(\frac{x}{6}\right) = \frac{1^3}{6} + \frac{2^3}{6} + \frac{3^3}{6} = \frac{1}{6} + \frac{8}{6} + \frac{27}{6} = \frac{36}{6} = 6$$

Example

What is the expected value of the continuous random variable X with pdf $f_x(x) = \frac{x+2}{18}$ for $-2 < x < 4$, 0 otherwise.

$$\Rightarrow E(X) = \int_{-2}^4 x \left(\frac{x+2}{18}\right) dx = \frac{1}{18} \int_{-2}^4 x^2 dx + \frac{2}{18} \int_{-2}^4 x dx = 2$$

Proof (For continuous case)

$\Rightarrow$ Suppose X is continuous. The proof is similar for discrete.

$\Rightarrow$ Replace integrals with sum

$$\begin{aligned} E(k_1g_1(x) + k_2g_2(x)) &= \int_{-\infty}^{\infty} (k_1g_1(x) + k_2g_2(x))f(x)dx = \int_{-\infty}^{\infty} k_1g_1(x)f(x)dx + \int_{-\infty}^{\infty} k_2g_2(x)f(x)dx \\ &= k_1 \int_{-\infty}^{\infty} g_1(x)f(x)dx + k_2 \int_{-\infty}^{\infty} g_2(x)f(x)dx \\ &= k_1 E(g_1(x)) + k_2 E(g_2(x)) \\ &= E(k_1g_1(x)) + E(k_2g_2(x)) \end{aligned}$$

both must be convergent (given by definition)